LAB 1

Requirements Engineering & UML Use-Case Modelling

Student / SRN	PES1UG24CS118
Project Title	Warehouse Inventory & Pallet Location Tracker
Primary Domain	Smart Cities, Transport & Logistics
Target Actors	Warehouse Operator, Logistics Supervisor

1. Scenario Overview

Project. Warehouse Inventory & Pallet Location Tracker.

What it does. An automated warehouse management system that keeps track of pallets in three-dimensional shelf positions, enforces each rack's maximum weight limit, records stock movements, and identifies pallets through barcode or RFID scanning.

Target actors. Warehouse Operator and Logistics Supervisor, as set in the brief.

Extra actor modelled. A Barcode/RFID Scanner, treated as a device actor **-** the lab material allows a device to act as an actor.

What Lab 1 asks for. A requirements table (five FRs and two NFRs), a UML use-case diagram with at least three actors, five use cases and an include/extend relationship, and one detailed use-case flow with preconditions, postconditions, a main success scenario and at least one alternate flow.

2. Requirements Table

Five functional and two nonfunctional requirements. Every row has a testable description, a measurable pass/fail check, a short reason and a review note.

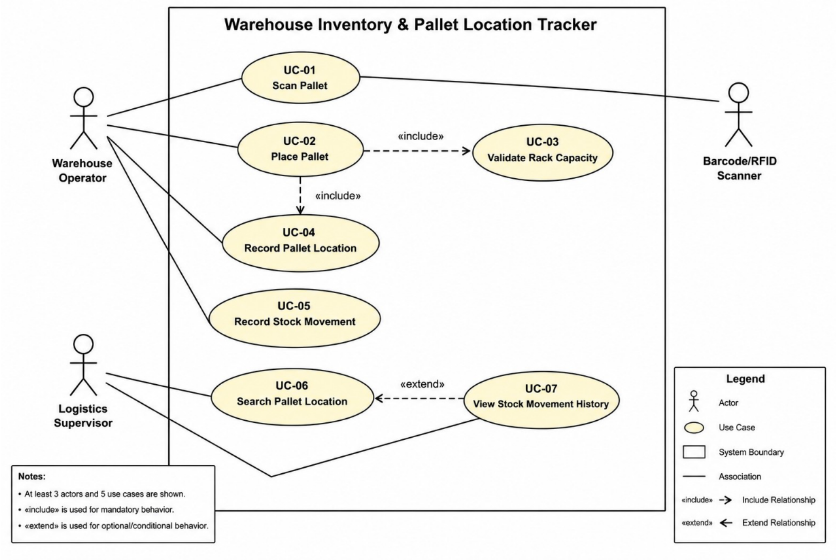
Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-001	Functional	The system shall check every proposed pallet position against the target rack's maximum load capacity before confirming storage, and reject any placement that would push the rack past its rated limit.	High	Pass - a pallet is confirmed only when the rack's resulting total weight stays within its rated structural limit. Fail - the system allows a placement that takes the rack over its rated limit.	Overloaded racks are a genuine safety hazard, so this guard protects both the staff and the building.	Given requirement - kept aligned with the scenario.
FR-002	Functional	The system shall look up a pallet from its barcode or RFID scan and return the matching inventory record.	High	Pass - a valid scan brings back the correct pallet record. Fail - an unknown or invalid tag is accepted as a real pallet.	Scanning is far quicker and less error-prone than typing identifiers by hand.	Covers the barcode/RFID identification part of the brief.
FR-003	Functional	The system shall store the full three-dimensional shelf position - warehouse, rack, shelf and slot - for every pallet it holds.	High	Pass - each stored pallet ends up with a complete, valid set of location coordinates. Fail - a pallet is saved with a missing or incomplete location.	Staff can only find stock quickly if the exact position is always on record.	Supports the 3D position-tracking requirement.
FR-004	Functional	The system shall log every pallet movement together with the pallet ID, source location, destination location, timestamp and the operator responsible.	High	Pass - a completed movement records all five details. Fail - a movement is saved with any of those fields missing.	A full movement history is what makes stock traceable when something goes wrong.	Gives a clear audit trail for inventory movements.
FR-005	Functional	The system shall let authorised users search for a pallet by SKU, barcode or RFID and show its current shelf coordinates.	Medium	Pass - a valid identifier returns the pallet's real current location. Fail - an invalid identifier returns a location for a pallet that does not exist.	Day to day, staff mostly need one thing fast: where is this pallet right now?	Everyday pallet lookup for the floor team.
NFR-001	Nonfunctional	The warehouse lookup service shall return any SKU's pallet coordinates across 100,000 bins in under 100 ms.	High	Pass - under simulated peak load, benchmarks show every lookup finishes in under 100 ms across the full 100,000-bin dataset. Fail - lookup time rises above 100 ms under that same peak load.	Lookups happen constantly, so even small delays add up quickly across a shift.	Given requirement - 100,000-bin / 100 ms target kept. Criteria now measures latency only; security sits with NFR-002.
NFR-002	Nonfunctional	The system shall limit inventory and stock-movement actions to signed-in users, according to the role each user has been given.	High	Pass - a signed-in user can only carry out the actions their role allows. Fail - an unauthenticated or wrong-role user is able to change inventory or movement records.	Keeps the inventory data trustworthy by stopping unauthorised edits.	Role-based access control.

3. UML Use-Case Diagram

Three actors and seven use cases. «include» is used for behaviour that always runs; «extend» for optional behaviour. Key links: UC-02 «include» UC-03, UC-02 «include» UC-04, and UC-07 «extend» UC-06.

3. UML Use-Case Diagram

The diagram models the three actors and seven use cases for the warehouse tracker. It uses «include» for mandatory behaviour and «extend» for optional/conditional behaviour.



Key relationships: UC-02 Place Pallet «include» UC-03 Validate Rack Capacity; UC-02 Place Pallet «include» UC-04 Record Pallet Location; UC-07 View Stock Movement History «extend» UC-06 Search Pallet Location.

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4. Use-Case Flow - UC-02 Place Pallet

Primary Actor: Warehouse Operator

Preconditions

- The Warehouse Operator is signed in.
- The pallet has a valid barcode or RFID tag.
- The pallet's inventory details already exist in the system.
- A proposed rack and shelf position has been chosen.

Postconditions

- The pallet is assigned to a valid 3D shelf position.
- The pallet's location is saved in the system.
- The rack's load information is brought up to date.
- The stock movement is recorded.

Main Success Scenario

1. The Warehouse Operator scans the pallet's barcode or RFID tag.
2. The system identifies the pallet and pulls up its inventory information.
3. The operator picks a proposed rack and shelf position.
4. The system reads the rack's current load and its maximum permitted load.
5. The system checks the proposed placement against that maximum load capacity.
6. The system confirms the placement is within limits.
7. The system saves the pallet's 3D rack, shelf and slot coordinates.
8. The system logs the movement with the pallet ID, source and destination, timestamp and operator.
9. The system confirms the pallet has been placed successfully.

Alternate Flow - 5a. Rack Capacity Exceeded

5a1. The system works out that this placement would take the rack over its maximum rated weight.

5a2. The system rejects the proposed placement.

5a3. The system tells the operator that the chosen rack cannot take the pallet.

5a4. The operator picks a different, valid rack and shelf position.

5a5. The use case picks back up from Step 4.